

Table 1: LSDA improvement over baselines for configuration dependency *identification*.

System	cDep	GPTuner	CoT	LLM-RAG
	Mean (improvement)	Mean (improvement)	Mean (improvement)	Mean (improvement)
Precision				
HDFS	0.125 (178.4%)	0.048 (625.0%)	0.097 (258.8%)	0.235 (48.1%)
HBASE	0.094 (351.1%)	0.029 (1362.1%)	0.059 (618.6%)	0.203 (108.9%)
YARN	0.406 (34.0%)	0.086 (532.6%)	0.111 (390.1%)	0.250 (117.6%)
MAPREDUCE	0.141 (175.9%)	0.062 (527.4%)	0.147 (164.6%)	0.283 (37.5%)
CORE	0.110 (391.8%)	0.036 (1402.8%)	0.073 (641.1%)	0.243 (122.6%)
GLANCE	—	0.075 (532.0%)	0.095 (398.9%)	0.303 (56.4%)
CINDER	—	0.056 (548.2%)	0.037 (881.1%)	0.091 (298.9%)
Average	0.175 (151.4%)	0.056 (685.7%)	0.088 (400.0%)	0.230 (91.3%)
Recall				
HDFS	0.519 (27.9%)	0.429 (54.8%)	0.729 (-8.9%)	0.369 (79.9%)
HBASE	0.455 (90.8%)	0.460 (88.7%)	0.369 (135.2%)	0.598 (45.2%)
YARN	0.608 (54.6%)	0.354 (165.5%)	0.607 (54.9%)	0.472 (99.2%)
MAPREDUCE	0.422 (110.9%)	0.244 (264.8%)	0.530 (67.9%)	0.419 (112.4%)
CORE	0.155 (427.7%)	0.080 (922.5%)	0.553 (47.9%)	0.541 (51.2%)
GLANCE	—	0.064 (1326.6%)	0.155 (489.0%)	0.268 (240.7%)
CINDER	—	0.090 (887.8%)	0.245 (262.9%)	0.218 (307.8%)
Average	0.432 (97.9%)	0.246 (247.6%)	0.455 (87.9%)	0.412 (107.5%)
F1-score				
HDFS	0.202 (125.7%)	0.086 (430.2%)	0.170 (168.2%)	0.287 (58.9%)
HBASE	0.156 (237.8%)	0.055 (858.2%)	0.101 (421.8%)	0.300 (75.7%)
YARN	0.487 (41.3%)	0.138 (398.6%)	0.187 (267.9%)	0.325 (111.7%)
MAPREDUCE	0.211 (155.5%)	0.098 (450.0%)	0.230 (134.3%)	0.334 (61.4%)
CORE	0.129 (400.8%)	0.049 (1218.4%)	0.129 (400.8%)	0.334 (93.4%)
GLANCE	—	0.069 (800.0%)	0.117 (430.8%)	0.281 (121.0%)
CINDER	—	0.069 (643.5%)	0.064 (701.6%)	0.127 (303.9%)
Average	0.237 (140.5%)	0.081 (603.7%)	0.143 (298.6%)	0.284 (100.7%)
AUC				
HDFS	0.758 (9.8%)	0.711 (17.0%)	0.862 (-3.5%)	0.684 (21.6%)
HBASE	0.725 (28.7%)	0.722 (29.2%)	0.681 (37.0%)	0.798 (16.9%)
YARN	0.803 (20.8%)	0.674 (43.9%)	0.800 (21.2%)	0.735 (32.0%)
MAPREDUCE	0.708 (33.3%)	0.618 (52.8%)	0.762 (23.9%)	0.708 (33.3%)
CORE	0.577 (57.5%)	0.538 (69.0%)	0.773 (17.6%)	0.769 (18.2%)
GLANCE	—	0.531 (80.0%)	0.577 (65.7%)	0.633 (51.0%)
CINDER	—	0.544 (73.5%)	0.617 (53.0%)	0.607 (55.5%)
Average	0.714 (29.8%)	0.620 (49.5%)	0.725 (27.9%)	0.705 (31.5%)